

Hitesh Bhatnagar

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Education

Vellore Institute of Technology

Sep 2022 – April 2026

Bachelor of Technology in Electronics and Communication Engineering; CGPA: 8.42/10

Vellore, Tamil Nadu

Experience

IT Risk Advisory Associate, Aumyaa Consulting Services LLP

Feb 2026 – Present

ITGC Audit & AI Solutions Development

Hybrid, India

- Conduct **IT General Controls (ITGC)**, **Audit Trail**, and **Information Systems Audits** across Access Management, Change Management, and IT Operations for BMW Financial Services, ONGC, and Jamnagar-based engagements.
- Perform **User Access Review (UAR)**, **Segregation of Duties (SOD)**, provisioning approval, privileged-access, and SAP transport testing.
- Prepare **Risk Control Matrices (RCMs)**, audit testing sheets, Management Letters, and Conclusion Memoranda with evidence mapping and control conclusions.
- Developed an internal **Python data anonymization tool** using OCR, NER, regex, and computer vision to detect and mask PII across PDF, Word, Excel, and image files.

Research Intern, Indian Institute of Technology Goa

May 2025 – Jul 2025

Deep Learning & Audio Processing

Goa, India

- Achieved **MSE 2.9e-5**, **RMSE 0.00537**, and **SNR 27.10 dB** on a hybrid Active Noise Cancellation model combining FxEHCAF and SFANC.
- Implemented adaptive learning-rate scheduling, momentum updates, and spectral weight decay in **Python** and **PyTorch** to improve convergence stability.
- Designed a lightweight **CNN** for noise classification and dynamic filter switching; optimized the training pipeline for CPU-based execution.

Publications

- Design of Logistic Distance Metric Based Robust Adaptive Filter for Active Noise Control** - *Digital Signal Processing*, Elsevier, 2026.
- Graph-Based Zero-Day IoT Botnet Detection** - *NEleX 2026*, ANRF-Sponsored, VIT, India, 2026.
- A Novel Three-Layer Hybrid Cryptographic Framework using Enhanced Classical Ciphers** - *IEEE ICAAIC*, 2025.

Projects

Jarvis C++ Inference Engine | C++17, llama.cpp, CMake, Docker, Linux

2025

- Built a **local LLM inference application in C++17** using llama.cpp, implementing tokenization, Top-K and temperature sampling, and context management.
- Added local **.txt** knowledge injection and restricted file I/O, with generated output confined to a Docker-mapped directory.
- Containerized the application with **CMake and Docker** for one-command execution with mounted GGUF models and host volumes.

Embedded RTOS Simulator | C, Linux, Systems Programming, GitHub Actions

2025

- Developed an **RTOS simulator in C** with priority scheduling, round-robin tie-breaking, sleep handling, and task-state management.
- Implemented message queues, binary semaphores, UART circular buffers, interrupt simulation, and a CLI shell.
- Added timer-driven updates, runtime diagnostics, queue and tick statistics, CPU-usage monitoring, and automated GitHub Actions builds.

Mini Database Engine in C | C, Data Structures, DBMS Internals

2024

- Built an **in-memory database engine in C** with a REPL-based SQL-like interface supporting insert, select, and delete operations.
- Implemented command parsing, fixed-size row storage, execution result handling, and input-length validation.
- Organized the database using result codes and a modular command-execution pipeline written in portable standard C.

Technical Skills

Languages: C, C++, Python, Java, Rust, SQL, Bash, MATLAB, Verilog

AI/ML & Signal Processing: PyTorch, TensorFlow, Keras, scikit-learn, CNNs, OCR, NER, RAG, LLM Inference, DSP

Systems & Developer Tools: Linux, Docker, CMake, Git, GitHub Actions, llama.cpp, AWS

Core Computer Science: Data Structures and Algorithms, DBMS, Computer Networks, Cryptography

IT Audit & Security: ITGC, IS Audit, SAP, Access Management, Change Management, UAR, SOD, RCM, PII Masking

Embedded & EDA: Embedded C, 8051, Keil uVision, Vivado, ModelSim, Proteus, Cadence